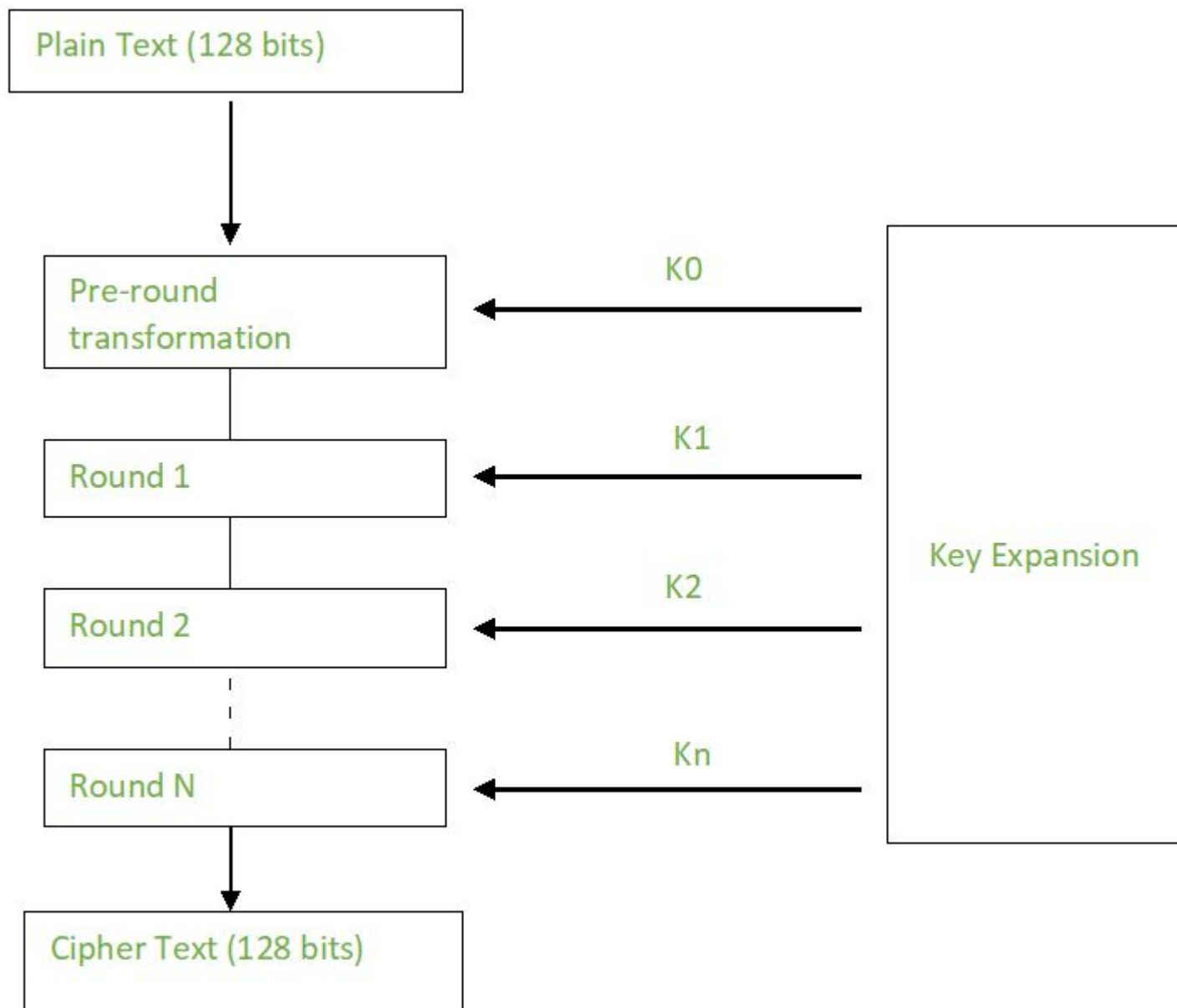
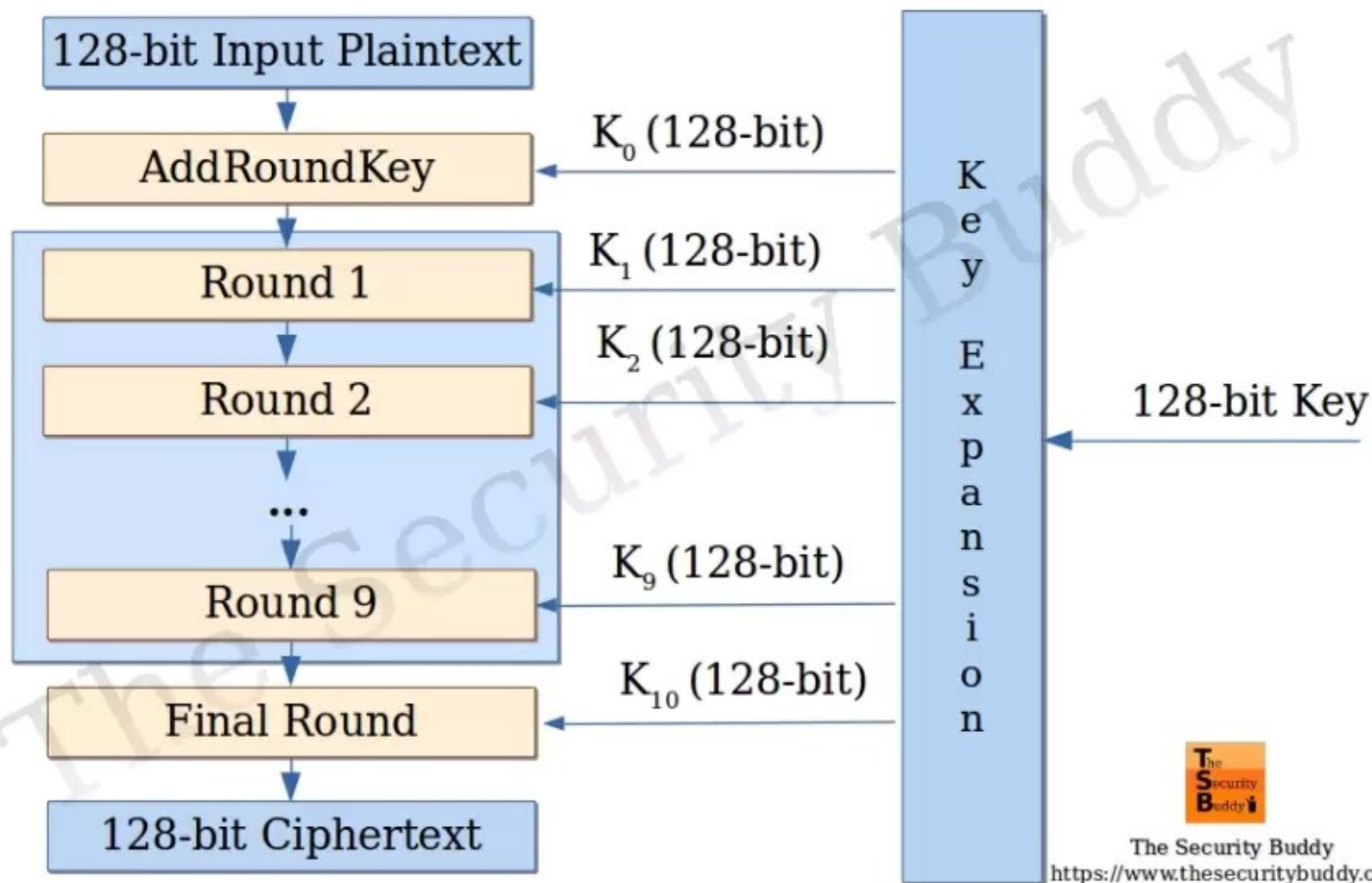


Advanced Encryption Standard (AES)

- AES is a [Block Cipher](#).
- The key size can be 128/192/256 bits.
- Encrypts data in blocks of 128 bits each.

N (Number of Rounds)	Key Size (in bits)
10	128
12	192
14	256





Each round comprises of 4 steps :

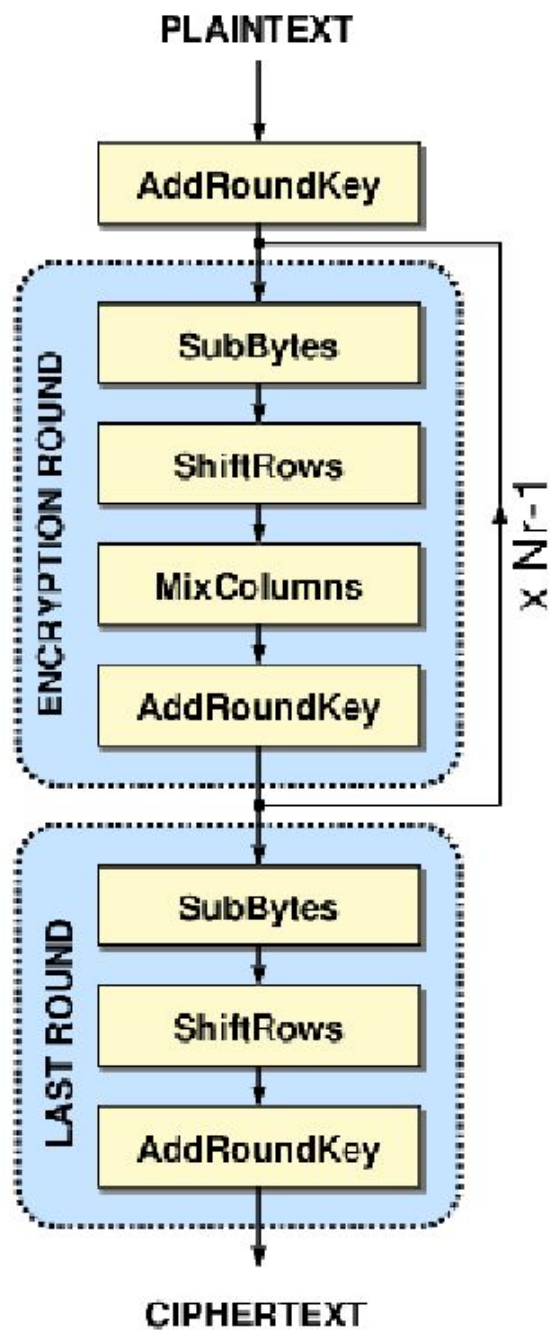
SubBytes

ShiftRows

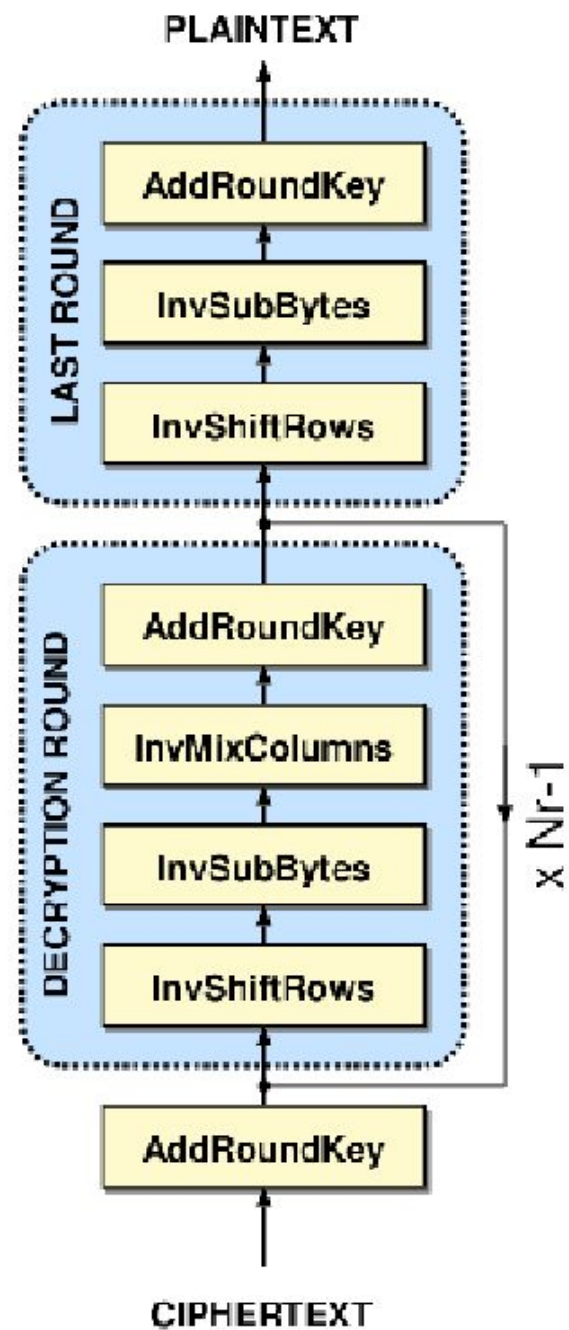
MixColumns

Add Round Key

ENCIPHERMENT



DECIPHERMENT



Step1. Sub Bytes

This step implements the substitution. In this step, each byte is substituted by another byte. It is performed using a lookup table also called the S-box.

Step2. Shift Rows

Each row is shifted a particular number of times.

The first row is not shifted

The second row is shifted once to the left.

The third row is shifted twice to the left.

The fourth row is shifted thrice to the left.

Step 3: Mix Columns

This step is a matrix multiplication.

Each column is multiplied with a specific matrix and thus the position of each byte in the column is changed as a result.

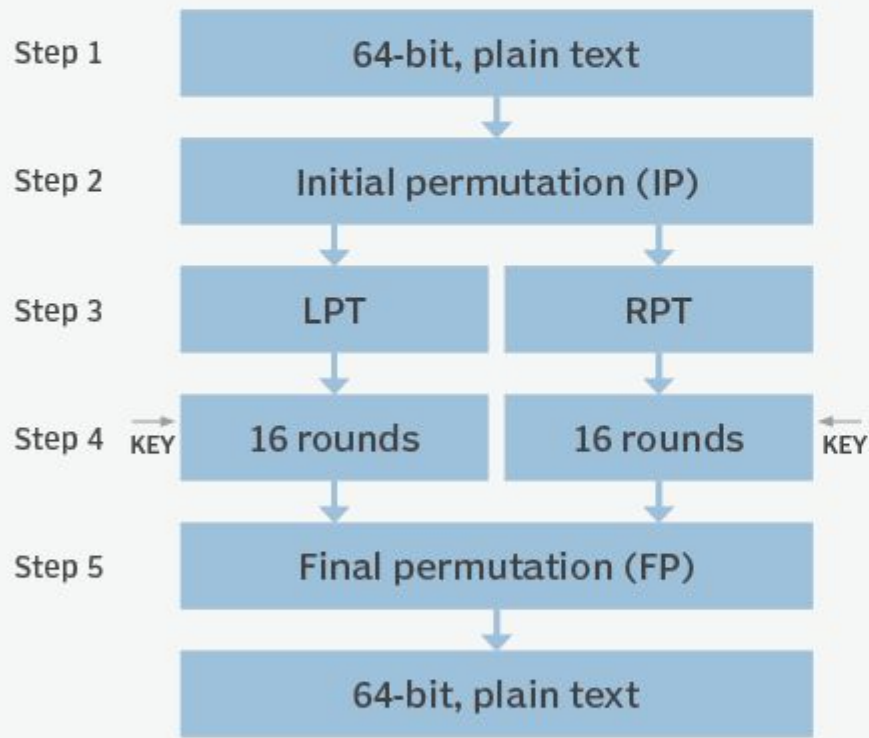
Step 4: Add Round Keys

Now the resultant output of the previous stage is XOR-ed with the corresponding round key.

After all these rounds 128 bits of encrypted data are given back as output. This process is repeated until all the data to be encrypted

DES encryption vs. AES encryption

DES encryption



AES encryption

